XINHAI HOU

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EDUCATION

University of Michigan, Ann Arbor, U.S.
Ph.D. in Bioinformatics and Scientific Computing | GPA: 4.0 — Sep 2021 - June 2026

- Advisor: **Prof. Todd Hollon, Prof. Honglak Lee**
- Research interests: computer vision, vision language model, agentic system

The Chinese University of Hong Kong, Shenzhen, China
B.S. in Statistical Science, First Class Honours (top 10%) — Sep 2016 - May 2020

PROFESSIONAL EXPERIENCE

Applied Scientist Intern | **Amazon** (Seattle, WA) — May 2025 - Aug 2025

- Led an end-to-end research project CodeV, addressing production bottlenecks in infographic extraction and multi-object counting/grounding, resulting in a paper accepted to **CVPR 2026** with **ORAL** presentation (1%).
- Post-train Qwen2.5-VL-7B for agentic visual reasoning on interleaved text, Python code, image with GRPO, boosting performance (by average 3.2%) and achieve SOTA on 10 visual perception and reasoning benchmarks.
- Identify and mitigate unfaithful tool usage by rubric-based reward and expert SFT pipeline (increase 30% faithfulness).

Research Assistant | **University of Michigan** (Ann Arbor, MI) — Sep 2021 - present

- Developed FastGlioma, a visual foundation model for fast (10x less) tumor detection. (**Nature, 2025**)
- Proposed Slide Pretraining Transformer (SPT) for gigapixel slide image embeddings. (**NeurIPS 2024, AIM-FM**)
- Introduced HiDisc, a hierarchical self-supervised learning framework for microscopy images. (**CVPR highlight, 2023**)
- Curated and released the first Stimulated Raman Histology dataset and benchmark. (**NeurIPS D&B track, 2022**)
- Built an agent harness for patient survival forecasting by constructing an evaluation pipeline from 10M+ medical reports, improving the C-index by 15% via a multi-agent system with self-calibration mechanisms.

Machine Learning Engineer | **Tencent** (Beijing, China), advised by Dr. **Pengfei Xiong** — Feb 2021 - Aug 2021

- Fine-tuned GPT-2 and UniLM on social media and news corpus for title generation used in WeChat Search.
- Doubled the training speed by distributed training and increased 5% in the ROUGE score by beam search.
- Won **5th** award out of 4335 participants in an ACM challenge: Multimodal Video Advertisement Competition.

SELECTED PUBLICATIONS

(* for co-first contribution)

CodeV: Code with Images for Faithful Visual Reasoning via Tool-Aware Policy Optimization
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026 (**Oral** presentation, top 1%)
Xinhai Hou, ..., Bryan Wang.

CORRECT: COndensed eRror RECognition via knowledge Transfer in multi-agent systems
International Conference on Machine Learning (**ICML**), 2026
Yifan Yu, ..., **Xinhai Hou**, ..., Bryan Wang

Foundation models for fast, label-free detection of glioma infiltration
Nature, 2025
Akhil Kondepudi, Melike Pekmezci, **Xinhai Hou**, ..., Todd Hollon.

A self-supervised framework for learning whole slide representations
Neural Information Processing Systems (**NeurIPS**) on AIM-FM, 2024
Xinhai Hou, ..., Honglak Lee, and Todd C. Hollon.

Hierarchical discriminative learning improves visual representations of biomedical microscopy
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023 (**Highlight**, top 2.5%)
Cheng Jiang*, **Xinhai Hou***, ..., Honglak Lee, and Todd C. Hollon.

OpenSRH: optimizing brain tumor surgery using intraoperative stimulated Raman histology
Conference on Neural Information Processing Systems (**NeurIPS**), 2022
Cheng Jiang*, **Xinhai Hou***, ..., Honglak Lee, and Todd C. Hollon.

Other publications:

Towards Scalable Language-Image Pre-training for 3D Medical Imaging
Transactions on Machine Learning Research (**TMLR**), 2026
Chenhui Zhao, ..., **Xinhai Hou**, ..., Todd C. Hollon

Learning neuroimaging models from health system-scale data
Nature Biomedical Engineering, 2026
Yiwei Lyu,..., **Xinhai Hou**, ..., Honglak Lee, and Todd C. Hollon

Step-calibrated diffusion for biomedical optical image restoration
AAAI Conference on Artificial Intelligence, 2025
Yiwei Lyu,..., **Xinhai Hou**, ..., Honglak Lee, and Todd C. Hollon

Super-resolution of biomedical volumes with 2D supervision
CVPR Workshop on Computer Vision for Microscopy Image Analysis, 2024
Cheng Jiang, ... **Xinhai Hou**, ..., Honglak Lee, and Todd C. Hollon

CORE COURSEWORK

Machine Learning	Computer Vision, Large Language Modeling, Machine Learning foundations
Statistics	Data Mining, Statistical Inference, Time Series, Nonparametric Statistics
Computer Science	Data Structure and Algorithm, Parallel Computing, Database Management
Bioinformatics	Bioinformatics concepts and algorithms, Biology for computational scientists

SKILLS

Programming languages	Python, Shell, C/C++
Framework	uv, Pytorch, vLLM, VeRL, OpenCV
Tools	Git, Vim, VS Code, AWS, LaTeX, Adobe Illustrator

AWARDS

- **MICDE Graduate Fellowship** 2024-2025
  Michigan Institute for Computational Discovery and Engineering, U of M
- **CVPR 2023 Scholarship** 2023
- **Dean's List** (Top 5%) 2016 - 2020
  The Chinese University of Hong Kong, Shenzhen
- **Undergraduate Research Award** 2016 - 2020
  The Chinese University of Hong Kong, Shenzhen

SERVICES

- **Conference reviewer / Program Committee**: CVPR 2024-26, NeurIPS 2022-25, ICML 2025, ICLR 2025, ECCV 2024
- **Journal Reviewer**: Journal of Digital Imaging, Journal Of Imaging Informatics In Medicine